AI Job Market Monitoring Dashboard

A comprehensive Power BI dashboard analyzing AI job demand, salaries, experience levels, and work types.

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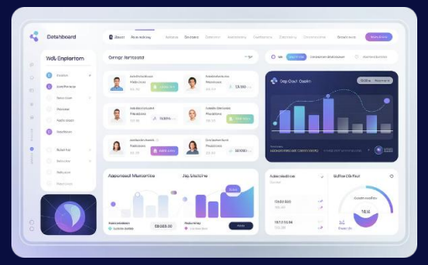
Project Scope & Objectives



Track Job Openings

Track AI vacancies

Analyze Salary Trends

Analyze compensation patterns

Evaluate Work Models

Assess work arrangements

Support Decisions

Provide data-driven insights

Identify Skill Requirements

Determine key AI skills

Track Company Hiring Patterns

Monitor recruitment trends

INTRODUCTION : Navigating the AI Frontier

Market Expansion

Rapid growth in the AI job market due to technological advancements.

Power BI Analysis

Power BI analyzes complex data for clear insights.

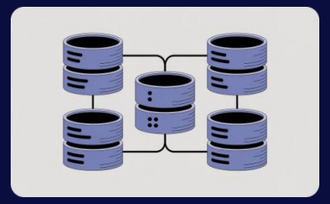
Actionable Insights

Provide actionable insights to help AI stakeholders make informed decisions.

Methodology: Power BI Dashboard Development

1. Data Import & Preparation

Import, clean, and transform data.



2. Data Modeling

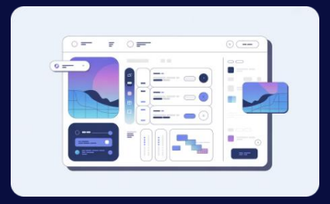
Model data, create calculated columns.

3. DAX Formulas for KPIs

Develop DAX for key KPIs.

4. Visualization Design

Design interactive charts and KPIs.



5. Interactive Slicers

Implement interactive filters.

6. Dashboard Layout

Design two-page dashboard layout.

Dataset

	A	B	C	D	E	F	G	H	I	J	K	L
1	job_id	company_name	industry	job_title	skills_required	experience_level	employment_type	location	salary_range_usd	posted_date	company_size	tools_preferred
2	1	Foster and Sons	Healthcare	Data Analyst	NumPy, Reinforcement Learning, PyTorch, Scikit-learn, GCP, FastAPI	Mid	Full-time	Tracybury, AR	92860–106588	20-08-2025	Large	KDB+, LangChain
3	2	Good, Myers and Ramirez	Tech	Computer Vision Engineer	Scikit-learn, CUDA, SQL, Pandas	Senior	Full-time	Lake Scott, OU	78523–144875	23-09-2024	Large	FastAPI, KDB+,
4	3	King Inc	Tech	Quant Researcher	MLflow, FastAPI, Azure, PyTorch, SQL, GCP	Entry	Full-time	East Paige, CM	124498–217204	18-09-2026	Large	BigQuery, PyTorch, Scikit-learn
5	4	Cooper, Archer and Lynch	Tech	AI Product Manager	Scikit-learn, C++, Pandas, LangChain, AWS, R	Mid	Full-time	Perezview, FI	60908–129743	09-08-2024	Large	TensorFlow, BigQuery, MLflow
6	5	Hall LLC	Finance	Data Scientist	Excel, Keras, SQL, Hugging Face	Senior	Contract	North Desireeland, NE	98884–156413	24-02-2026	Large	PyTorch, LangChain
7	6	Ellis PLC	E-commerce	AI Product Manager	GCP, Excel, Scikit-learn, MLflow	Senior	Remote	South Kevin, TZ	92852–180718	07-08-2025	Large	PyTorch, TensorFlow,
8	7	Martinez-Moore	Automotive	Data Analyst	Python, Flask, Scikit-learn, GCP	Mid	Full-time	West Shawn, LR	70578–80534	23-12-2024	Large	PyTorch, MLflow
9	8	Mullins Ltd	Education	Data Scientist	Excel, SQL, Flask	Entry	Internship	Port Hailey, RU	69032–82444	14-08-2024	Large	BigQuery
10	9	Aguilar PLC	Healthcare	NLP Engineer	Pandas, LangChain, GCP, Azure	Entry	Internship	Gutermouth, GB	134239–188904	15-09-2025	Large	Scikit-learn, LangChain,
11	10	Parks LLC	Automotive	Computer Vision Engineer	Excel, Reinforcement Learning, TensorFlow	Senior	Full-time	Nicolashire, BA	79361–165029	07-08-2025	Large	MLflow, TensorFlow
12	11	Vaughan-Ortiz	Tech	Data Scientist	Keras, MLflow, Reinforcement	Entry	Internship	New Angelatown, LU	89084–129871	10-01-2024	Large	LangChain, BigQuery,
13	12	Johnson, Guzman and Hernandez	Finance	ML Engineer	Excel, Reinforcement Learning, Keras, Scikit-learn	Entry	Full-time	Joychester, LT	63989–138670	15-03-2024	Large	TensorFlow, PyTorch, BigQuery
14	13	Martin and Sons	E-commerce	NLP Engineer	C++, Hugging Face, Python, FastAPI	Entry	Full-time	South Michael, FM	102770–117020	27-09-2023	Startup	LangChain
15	14	Frank-Owens	Retail	Quant Researcher	Excel, MLflow, NumPy, Hugging Face, LangChain, FastAPI	Senior	Full-time	Christenberg, GQ	137004–158825	20-11-2024	Mid	LangChain, TensorFlow, Scikit-learn
16	15	Rogers Ltd	Education	NLP Engineer	Pandas, CUDA, NumPy, FastAPI	Mid	Remote	New Katisborough, FJ	128042–216741	08-07-2024	Large	FastAPI
17	16	Wright-Evans	Automotive	NLP Engineer	Hugging Face, Scikit-learn, Reinforcement Learning	Senior	Remote	Stephaniechester, LU	120869–217868	09-04-2024	Startup	TensorFlow
18	17	Preston Ltd	E-commerce	ML Engineer	MLflow, R, C++	Entry	Full-time	South Tim, CZ	137523–180097	14-06-2023	Mid	LangChain
19	18	Smith-Moore	Automotive	AI Product Manager	Excel, MLflow, Hugging Face, Flask, Power BI	Entry	Internship	Blevinsburgh, VU	133184–182657	29-04-2024	Startup	BigQuery
20	19	Watts and Sons	Healthcare	Data Scientist	AWS, Azure, Power BI, CUDA, Reinforcement Learning, Hugging	Senior	Remote	Patsmouth, CA	100845–188745	19-10-2024	Mid	BigQuery, FastAPI, KDB+
21	20	Gates LLC	E-commerce	ML Engineer	R, CUDA, Power BI, AWS	Mid	Internship	Steuenhaven, BW	116578–207180	28-08-2024	Large	MLflow
22	21	Whitaker, Walker and Sandoval	Retail	Quant Researcher	C++, Power BI, R, SQL, Keras, Azure	Entry	Remote	Gregorymouth, LV	122521–175188	18-01-2024	Large	Scikit-learn, BigQuery, PyTorch
23	22	Humphrey LLC	Retail	Quant Researcher	LangChain, NumPy, CUDA, C++, MLflow, R	Entry	Full-time	Lake Alexander, RU	62489–105216	25-12-2023	Large	PyTorch, LangChain
24	23	Adams, Stevenson and Wagner	E-commerce	NLP Engineer	CUDA, FastAPI, Scikit-learn, MLflow, Hugging Face	Mid	Remote	North Nicholas, SN	74317–148959	25-11-2024	Startup	PyTorch, Scikit-learn, KDB+
25	24	Thomas, Hurst and Becker	Tech	NLP Engineer	PyTorch, CUDA, Hugging Face, Flask, C++	Senior	Internship	Leston, MW	88116–126328	08-12-2023	Mid	Scikit-learn
26	25	Sherman-Nguyen	Healthcare	NLP Engineer	Python, Reinforcement Learning, MLflow	Mid	Internship	Susannaclev, TW	88640–108633	18-08-2025	Mid	FastAPI
27	26	Johnson-Keller	Finance	NLP Engineer	PyTorch, Reinforcement Learning, FastAPI, Power BI, Python, R	Mid	Remote	Whitestad, BB	107786–128713	20-12-2024	Mid	FastAPI, PyTorch, KDB+

Dataset Description

Key Attributes

Company, job title, experience, employment, location, salary.

Experience Levels

Entry, Mid, Senior levels (AI/data).

Work Types

Full-time, Contract, Remote, Internships.

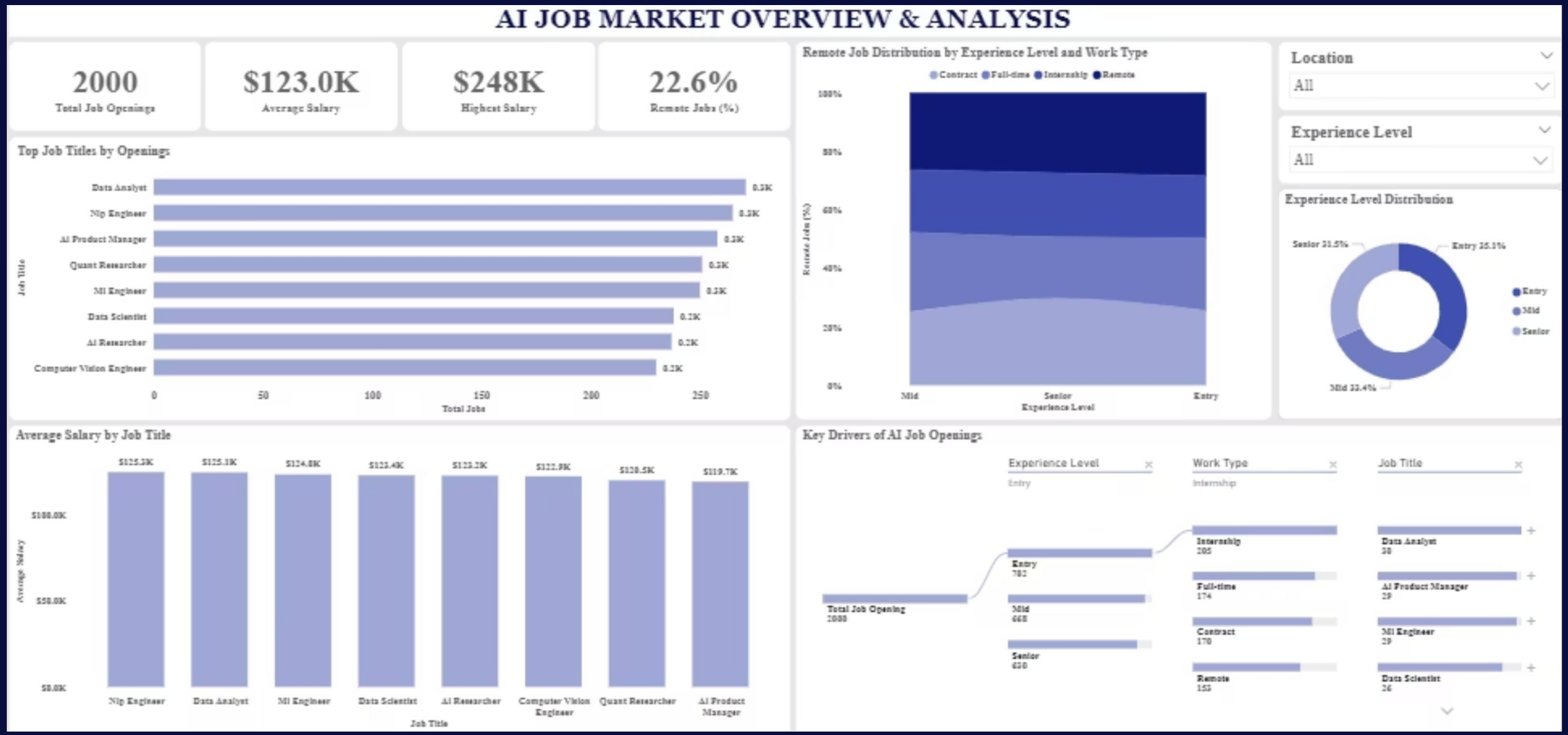
Salary Range

Compensation for all levels; avg. $123K.

Geographic Coverage

Job openings: multiple locations and remote.

Dashboard – Key Performance Indicators



Dashboard Components & Features

KPI Cards

Key metrics: Total Openings, Avg Salary, Remote.

Top Job Titles

AI job demand by horizontal bar chart.

Average Salary

Average salary by role (vertical bar chart).

Jobs by Company

Job distribution by company (stacked area chart).

Experience Distribution

Experience level proportions (donut chart).

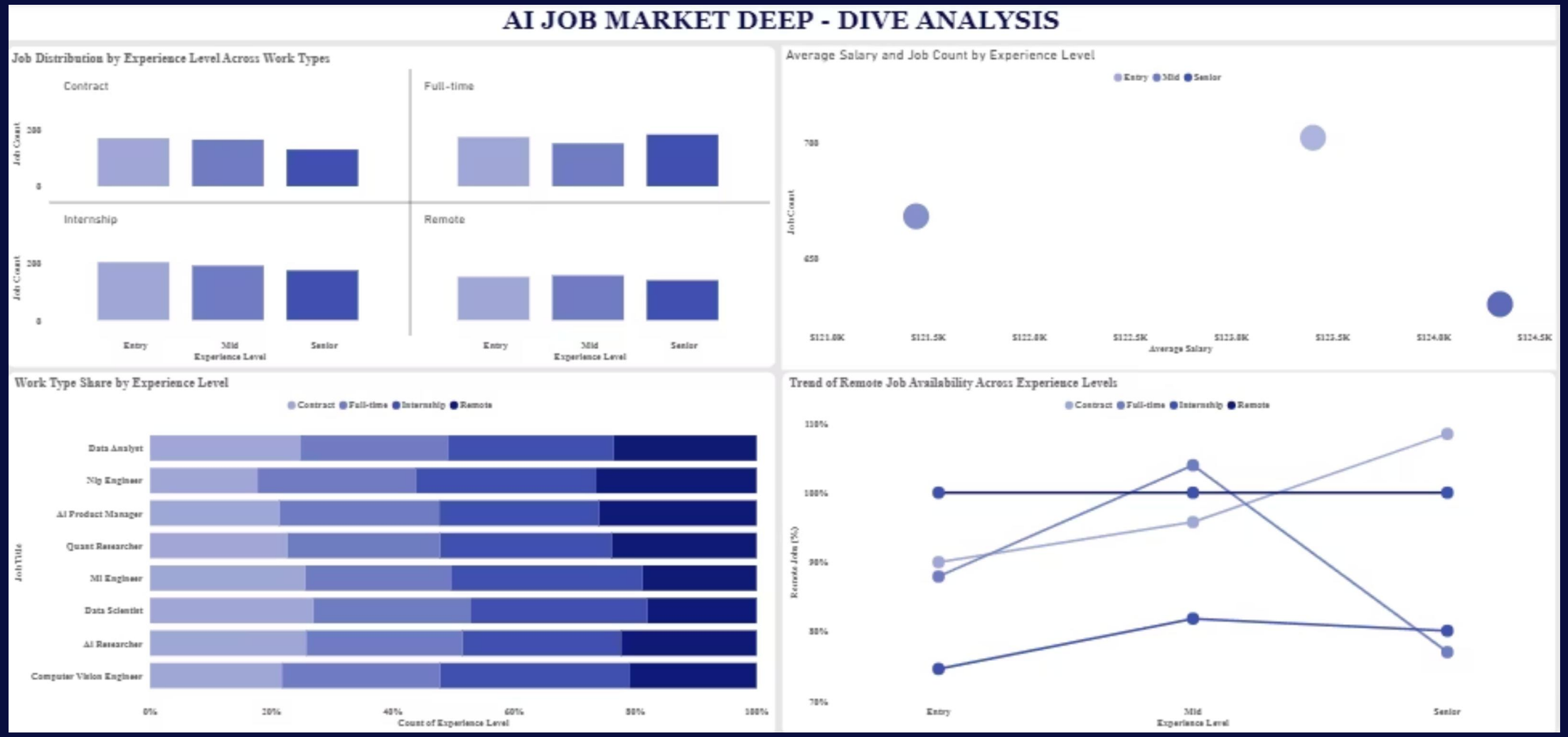
Experience Level Stats

Entry, Mid, Senior stats breakdown.

Interactive Slicers

Location & Experience filters.

Dashboard – Deeper Dive into Dynamics



Advanced Analytics & Filtering

Job Distribution

Counts by experience & work type.

Top Job Titles

Rankings by experience level.

Remote Job Count

Opportunities by experience level.

Remote Job Trend

Across experience levels, over time.

Key Insights : Unlocking the AI Market Landscape

Mid-level Dominance

Mid-level roles dominate AI job openings, indicating strong demand for experienced professionals.

High-Demand Roles

Data Analyst and Machine Learning Engineer are highly sought-after, crucial for data interpretation and AI development.

Senior Salary Premium

Senior-level roles command higher salaries, valuing extensive experience and specialized expertise.

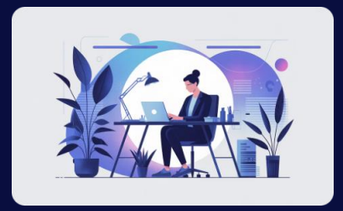
Remote Work & Experience

Remote work increases significantly with experience for seasoned professionals.

Strategic Recommendations: Navigating Your AI Career

Focus Skill Development: Prioritize mid-level AI skills for abundant opportunities

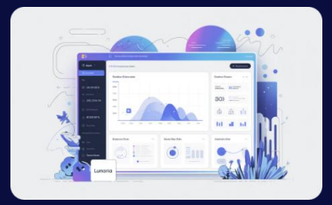
Upskill for Advanced Roles: Specialize in ML and AI engineering for high demand and competitive salaries



Target Remote-Friendly Roles: Seek remote opportunities as experience grows

Embrace Continuous Learning: Stay current with evolving AI technologies for competitiveness

Tools & Technologies: The Power Behind Our Insights



Power BI Desktop

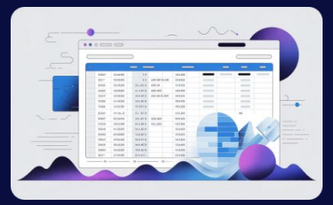
Primary for data visualization and dashboards.

DAX

For advanced calculations and KPIs.

CSV Dataset

Foundational data source.



Microsoft Excel

For initial data cleaning and preparation.

Thank You

I appreciate your time and attention today.